Test workflow

1. Upload test data
2. Explore measurements
3. Review engineering metrics
4. Investigate events
5. Export reports
6. Compare two tests

Reset analysis settings

Thermal Test Intelligence Center

Upload → Document → Visualize → Analyze → Review → Export | TTIC 1.0 · Stage 2D.3

Upload a test file

Sample Data.xlsx
1.8MB

Worksheet

Sheet1

Timestamp column

Timestamp

Observations	Numeric channels	Duration	Median sample interval
6,899	59	11.0 min	0.094 s

① Test Setup ② Visualization ③ Engineering Analytics ④ Event Intelligence ⑤ Reports & Export ⑥ Test Comparison

Compare any two thermal test files

Stage 2C: compare full recordings, equal elapsed durations, or independently selected test phases. No interpolation, pairing, or automatic pass/fail judgment.

Test A — CSV or Excel

Sample Data.xlsx
1.8MB

Test B — CSV or Excel

7478_GE...072026.csv
3.3MB

A worksheet

Sheet1

B timestamp column

Timestamp

A timestamp column

Timestamp

Test A measurement

TH1

Test B measurement

TH1

Comparability check: verify that selected channels measure the same physical quantity, use compatible units, and have comparable test conditions. A numerical difference alone does not establish improvement or failure.

Both tests retain their original readings and sampling rates. Time windows filter raw records; the chart may display fewer points for responsiveness.

Test A

6,899 samples

↑ 10.97 min

Test B

6,899 samples

↑ 10.97 min

Comparison time windows

Comparison mode

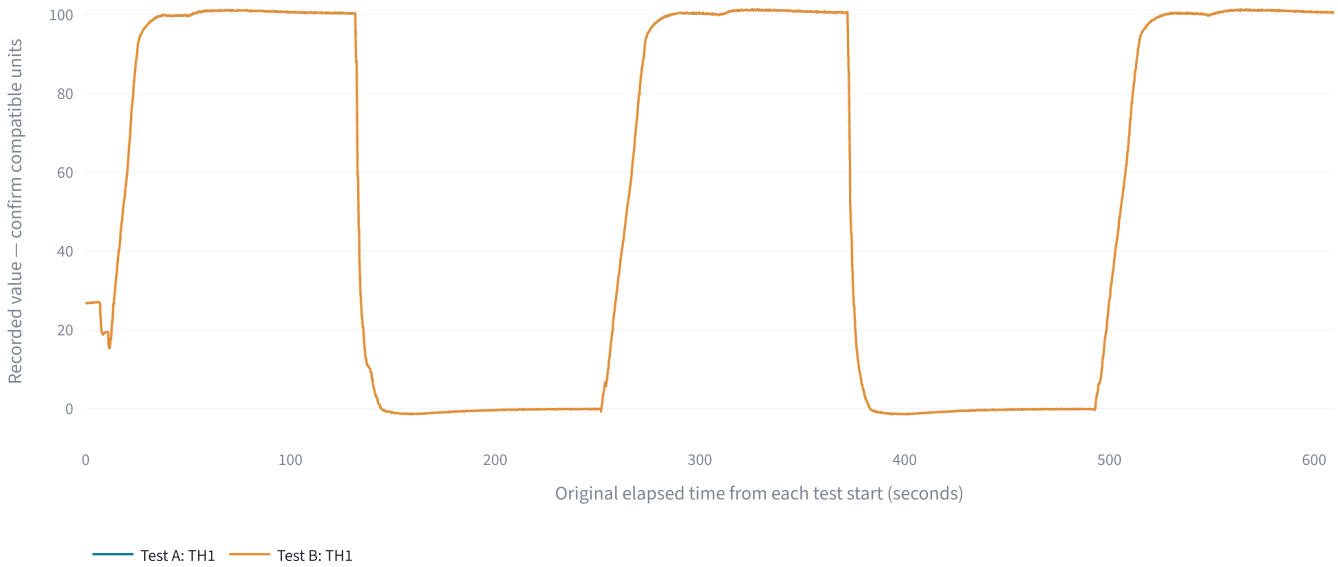
Full Test | Common Duration | Independent Windows

Align selected window starts at zero on the chart

Test A: 0.000–658.062 s, 6,899 valid readings. Test B: 0.000–658.062 s, 6,899 valid readings.

Approximate displayed points per test

4000



Download comparison overlay (HTML)

Independent test statistics · selected windows

Metric	Test A	Test B	Difference (B − A)	Change (%) vs |A|
Samples	6899	6899	0	None
Minimum	-1.3771	-1.3771	0	0
Maximum	101.3454	101.3454	0	0
Mean	53.098	53.098	0	0
Std dev (sample)	46.7869	46.7869	0	0
Range	102.7225	102.7225	0	0

All metrics use valid, unsmoothed raw readings within each selected window, not chart samples. Percentage change uses |A| as the denominator; it is omitted for sample counts and zero baselines. No readings are interpolated or paired.

Download comparison metrics (CSV)

Heating & cooling rate intelligence · Stage 2D.1

Calculate raw-reading rates within each selected test window. This is a screening comparison, not automatic identification of physical heating/cooling phases.

☐ Both selected channels are temperature in °C

Rate calculation interval (seconds)

30.00 − +

Metric	Test A	Test B	Difference (B − A)
Samples	6899	6899	0
Elapsed span (s)	658.062	658.062	0
Endpoint change	0.376	0.376	0
Overall endpoint rate (recorded units/min)	0.0343	0.0343	0
Max heating rate (recorded units/min)	158.1884	158.1884	0
Max cooling rate (recorded units/min)	-204.3483	-204.3483	0
Eligible intervals	22	22	0

Overall endpoint rate = (last − first valid reading) / elapsed time × 60. Interval rates use the first and last recorded values in each nonoverlapping interval; maximum heating is the largest positive interval rate and maximum cooling is the most negative. A missing heating/cooling rate means no eligible interval showed that direction, not zero rate. No smoothing, interpolation, or cross-test sample pairing is performed.



Download rate comparison summary (CSV)

Download Test A interval rates (CSV)

Download Test B interval rates (CSV)

Steady-state screening · Stage 2D.2

Identify low-variation periods within the selected comparison windows. This is a screening classification, not proof of thermal equilibrium.

Stability window (seconds) 30.00 − +

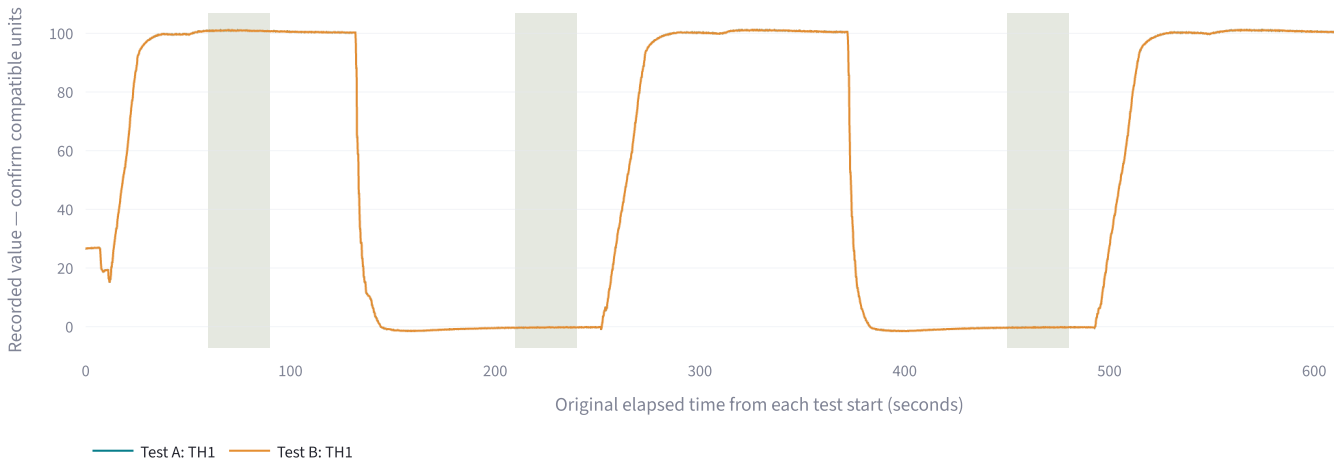
Maximum variation (recorded units) 0.50 − +

Minimum unique readings per window 5 − +

A window qualifies when max − min ≤ tolerance, it contains enough unique readings, and observations span at least 80% of the window. Incomplete final windows and insufficiently sampled windows are not classified as stable.

Metric	Test A	Test B	Difference (B − A)
Eligible windows	21	21	0
Stable windows	3	3	0
Stable window fraction (%)	14.2857	14.2857	0
Detected periods	3	3	0
Longest stable period (s)	30	30	0
Total qualified window time (s)	90	90	0

Stable window fraction is calculated from eligible windows only. Total qualified time counts complete stable windows, not the total test duration. Adjacent stable windows are grouped; the overall variation of a grouped period may exceed the per-window tolerance.



Shaded regions indicate detected stable periods (teal: Test A; orange: Test B). Overlapping regions may visually blend. No cross-test time alignment beyond the chart setting is inferred.

Detected stable periods

Test A — 3 periods

Start (s)	End (s)	Duration (s)	Stable windows	Readings (unique timestamps
60	90	30	1	317
210	240	30	1	308
450	480	30	1	318

Test B — 3 periods

Start (s)	End (s)	Duration (s)	Stable windows	Readings (unique timestamps)
60	90	30	1	317
210	240	30	1	308
450	480	30	1	318

Download Test A stable periods (CSV)

Download Test B stable periods (CSV)

Download Test A stability windows (CSV)

Download Test B stability windows (CSV)

Download stability comparison summary (CSV)

Command-to-response intelligence · Stage 2D.3

Detect command steps and measure the selected sensor response. Results are observational screening estimates, not certified response times.

Test A command channel: TSET1

Test B command channel: TSET1

Test A response channel: TH1

Test B response channel: TH1

Minimum command step (units): 5.00

Merge nearby events within (s): 2.00

Pre-event baseline (s): 10.00

Post-event observation (s): 60.00

Minimum measurable response change (recorded units): 0.50

Final-reference tail duration (s): 10.00

Automatic events are detected from adjacent raw command readings. Closely spaced steps are consolidated; verify the selected event against the command trace.

Test A: 6 detected command events

Test A event selection: (•) Automatic event ( ) Manual event time

Test A detected event: 10.720 s · step +77

Event time (s)	Command before	Command after	Command step
10.72	23	100	7
130.777	100	0	-1(
250.905	0	100	1(
371.212	100	0	-1(
491.655	0	100	1(
611.874	100	23	-7

Download Test A command events (CSV)

Test B: 6 detected command events

Test B event selection: (•) Automatic event ( ) Manual event time

Test B detected event: 10.720 s · step +77

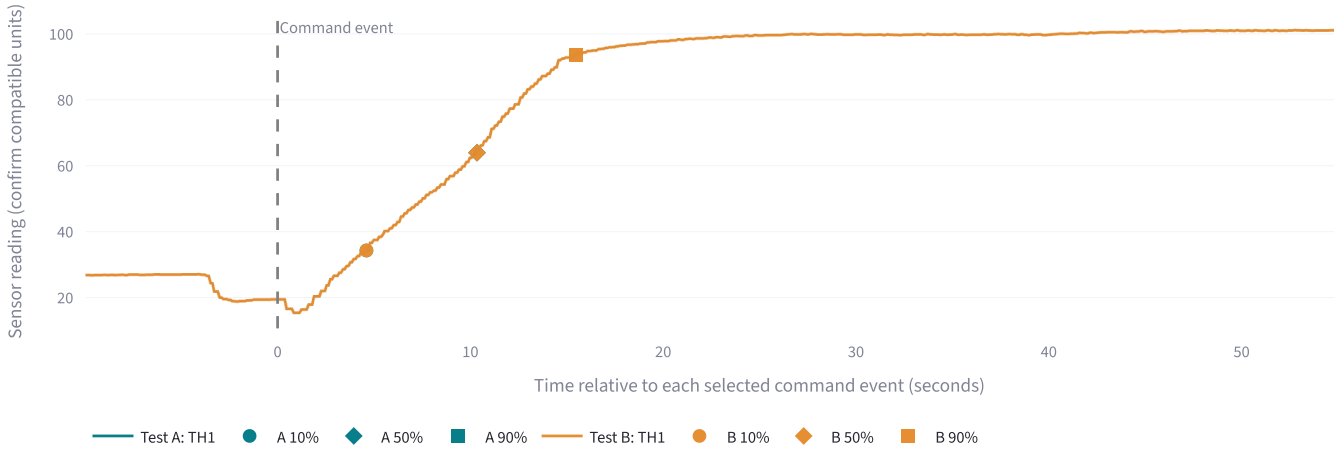
Event time (s)	Command before	Command after	Command step
10.72	23	100	77
130.777	100	0	-100
250.905	0	100	100
371.212	100	0	-100
491.655	0	100	100
611.874	100	23	-77

Download Test B command events (CSV)

Selected-event response metrics

Event time (s)	Status	Baseline	Final reference	Response change	t10 (s)	t50 (s)	t90 (s)	10–90 time (s)	Direction	Test	Command channel	Response ch
10.72	Measured	26.8587	101.0377	74.179	4.608	10.342	15.498	10.89	Rising	A	TSET1	TH1
10.72	Measured	26.8587	101.0377	74.179	4.608	10.342	15.498	10.89	Rising	B	TSET1	TH1

t10/t50/t90 are elapsed seconds after the command event to the first three-consecutive-reading threshold crossing. Baseline and final reference use medians; the latter is not proof of equilibrium. No interpolation is used. Overlapping command events are flagged rather than assigned a response time.



Download selected response metrics (CSV)

Engineering comparison reports · Stage 2D.4

Generate Excel and PDF from the same current comparison settings. Reports contain automated screening results and clearly separate engineer-entered judgments.

Report identification and engineering review

Report — Test A ID

Report — Test B ID

Report — Test A device/model

Report — Test B device/model

Report — Test A engineer

Report — Test B engineer

Report — Test A test type

Report — Test B test type

Comparability confirmation

Not verified

Comparison review status

Draft

Engineering observations (engineer entered)

Engineering conclusions (engineer entered)

Recommendations and follow-up

Prepare both comparison reports